# Biomedical Knowledge Graph Construction Using Large Language Models: Evaluating Generalisation Beyond Benchmark Datasets

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## 2.1 Biomedical Information Extraction

### 2.1.1 Definition

Biomedical IE is the process of automatically identifying structured information - entities such as genes, diseases, chemicals and species, and the relationships between them - from unstructured biomedical text such as journal abstracts and full-text articles. In this project, IE is treated as the first stage of a larger system: extracted entities and relations are later assembled into a Knowledge Graph (KG) and validated against biomedical ontologies **[NEED REFERENCE]**.

### 2.1.2 Motivation

The volume of biomedical literature has expanded at a rate that exceeds the capacity of manual review, with an analysis by Lu et al. (2011) demonstrating an exponential growth in the number of biomedical publications [1]. This growth is supported by more recent data - a 2024 study mapping over 20 million PubMed abstracts has demonstrated the magnitude of contemporary biomedical literature to a level where the volume of biomedical publications has made it difficult for researchers to keep track of new literature [2]. Literary analysis is not only a volume issue, but also an issue with discovery - log analysis of PubMed search behaviour shows that over 80% of views fall within the first page (top 20 results) of returned literature [3], meaning a substantial majority of literature goes largely unseen by researchers relying on manual review of ranked results.

Together, these findings motivate the need for automated IE for two reasons: the breadth of literature is growing faster than it can be analysed [1, 2] and a substantial majority of literature goes unseen that automated extraction and structuring could help mitigate [3]. This motivation supports the choice of PubMed as the source of the contemporary corpus for this thesis, as a corpus that is both large and continuously updated is exactly the setting in which static benchmark-trained pipelines are most likely to be used.

### 2.1.3 Tasks

Biomedical IE is usually comprised of several sub-tasks: Named Entity Recognition (NER), which identifies and stores types of spans of text referring to biomedical concepts, and Relation Extraction (RE), which identifies semantic relationships between recognised entities, and Named Entity Normalisation (NEN), which maps extracted mentions to pre-defined identifiers from an existing ontology [4, 5].

Unlike NER, which only labels a span of text as belonging to a given entity type, NEN maps that span to an established identifier from a pre-defined ontology or controlled vocabulary, storing which specific concept the span is referring to, rather than what kind of concept it is. This resolves terminology ambiguity challenges discussed in the following section.

### 2.1.4 Challenges

Long or complex sentences, which are common in biomedical abstracts, summarising multiple findings into a single sentence, can cause a significant challenge for these tasks [6]. This is particularly prominent for relation extraction, which typically requires attention over the text's distance between the two entities within a sentence: if two entities are far apart in this text, multiple clauses and mentions of competing entities can make it more difficult to correctly associate the entities with their corresponding relation [7], increasing the risk of both False Negatives (FNs) and False Positives (FPs). Sequential relation extraction models have been shown to have difficulty capturing these long-range dependencies, resulting in poorer performance on longer sentences [8]. The information density of biomedical literature may partly reflect conventions of scientific communication, which emphasise conciseness, clarity and precision of information [9].

Beyond task complexity, biomedical text includes domain-specific challenges that separate it from general IE: dense and evolving terminology [10, 11], and high rates of entity name ambiguity [12], where the same underlying concept is frequently expressed through several distinct strings, such as a drug's brand name versus its generic name, or a gene's full name versus a locally-defined abbreviation within a specific paper, while on the other hand, a single string can refer to different concepts depending on context. As a result, extracted entities referring to the same concept would be treated as distinct, fragmenting information than consolidating it. NEN aims to address this, mapping ambiguous or synonymous mentions onto a shared established identifier. This task is largely important for subsequent KG construction, as graph nodes rely on identifiers to be queryable and to allow separate mentions of the same concept to be merged into a single node **[NEED REFERENCE]**.

In addition to these domain-specific challenges, significant methodological limitations compound the difficulty of RE evaluation. Meaningful comparison between biomedical IE systems requires consistent evaluation frameworks: differences in annotation guidelines, entity and relation identifiers, and train/test splits across benchmark datasets can make performance difficult to compare directly between studies, even when the underlying literature appears homogeneous **[NEED REFERENCE]**. Separately, strong generalisation across literature distributions requires models to perform reliably across heterogeneous literature - literature differing in general publication date, subdomain, terminology, and reporting style from that of what a system was developed and evaluated on **[NEED REFERENCE]**. Benchmark datasets are fixed at the point of annotation, while biomedical literature continues to grow and evolve, therefore, strong performance on a static, in-distribution benchmark therefore does not offer any guarantee of comparable performance on literature published after the benchmark was constructed, or on subdomains that may have been underrepresented within it. This distinction between benchmark-reported and generalised performance is further discussed in Section 2.5, and motivates evaluating the frozen extraction pipeline developed against a separately constructed contemporary corpus.

### 2.1.5 Traditional Methods and Evaluation

Before generative LLMs became viable for biomedical IE, the field progressed through several distinct systems. Initial systems relied on manually-constructed rules and biomedical ontologies, matching text against predefined patterns and dictionaries to identify entities and relations **[NEED REFERENCE]**. These were gradually improved by and progressively replaced by traditional machine learning methods, which used syntactic and contextual features to train classifiers such as support vector machines for NER and RE **[NEED REFERENCE]**. Neural systems were then gradually implemented, initially through architectures such as recurrent and convolutional neural networks trained on word representations and later through transformer-based Pretrained Language Models (PLMs) fine-tuned specifically on biomedical text, such as BioBERT and PubMedBERT **[NEED REFERENCE]**. The progression of pretraining on biomedical corpora and task-specific fine-tuning allowed PLMs to substantially outperform earlier feature-based methods, resulting in them becoming the main extraction-system approach for specialised biomedical IE prior to later prompt-based generative approaches.

Development and evaluation throughout this progression have mainly been driven by benchmark datasets, with models commonly trained and evaluated on fixed, established corpora. However, relying on individual benchmarks introduces comparability and generalisability limitations outlined previously. Recent work has therefore aimed to address these limitations through more consistent evaluation frameworks and broader heterogeneous training data: Sanger et al. (2025) compared multiple PLMs before evaluating knowledge extraction of the best-performing model across five datasets within a consistent evaluation framework

[13]. Similarly, Lai et al. (2023) introduced BioREx, which consolidates heterogeneous biomedical RE datasets to support broader training and improve generalisation [14].

Both of these approaches, however, rely on PLMs tuned on labelled benchmark data, with performance assessed against fixed, in-distribution, annotated datasets. This highlights an important methodological limitation for practical application, where extraction systems must perform robustly on dynamic, contemporary literature, rather than a static benchmark corpus.

## 2.2 Large Language Models for Biomedical Information Extraction

### 2.2.1 Transition from Specialised Models to LLMs

More recently, LLMs have driven a broader shift in knowledge extraction and KG construction from rule-based and module pipelines towards generative and adaptive frameworks [15]. Early evidence supports competitive extraction performance with Zhang et al. (2024) reporting an F1 score of 0.84 for GPT-4 on biomedical RE for the Genetic Association Database (GAD) benchmark dataset, with GPT, BioBERT, and PubMedBERT performing comparably on the GAD dataset under certain extraction system frameworks [16]. However, it is important to note that GPT-3.5 and BioBERT v1.1 performed considerably worse than PubMedBERT on the masked ChemProt dataset, suggesting that relative model performance may vary substantially across biomedical RE tasks and datasets.

Additionally, performance gains from LLMs are not limited to biomedicine. Evaluations of LLMs for KG construction have observed an F1 score of 99.35% for Meta's LLaMA 3 model on OpenStack system log data [17], indicating extraction capability is not domain-specific.

This body of evidence motivates the pipeline design adopted in this project: an LLM-based extraction approach using structured JSON outputs and few-shot prompting, developed on an existing benchmark, BioRED [18], before evaluation on a newly-assembled contemporary corpus.

### 2.2.2 Prompt-Based and Few-Shot Extraction

Unlike specialised pretrained language models such as BioBERT and PubMedBERT, which require task-specific fine-tuning on labelled data to perform NER or RE, generative LLMs can perform biomedical IE directly through prompting, without any parameter tuning [19]. This approach can allow flexibility through natural-language instructions, rather than requiring additional model training. As a result, a single general-purpose model can potentially be adapted to different entity types, relation schemas, and extraction tasks through only prompt refinement

**[NEED REFERENCE]**.

Several approaches can be implemented: zero-shot prompting - where the model is given only a natural-language description of the task, few-shot prompting - where the prompt includes structured examples of exemplary outputs corresponding to given inputs, and schema-guided instruction, where the prompt specifies the exact entity and relation types that the model should extract [20, 21, 22].

The extent to which an LLM is able to compete with fine-tuned extraction systems depends heavily on how the prompt itself is constructed. Recent comparisons between zero-shot and few-shot prompting on biomedical IE tasks suggest that including examples in the prompt can improve extraction quality, as can annotation guidelines or provided schema within the prompt itself [21]. However, these gains are not consistent across biomedical IE tasks: Zhang et al. (2024) observed only limited improvements from one-shot prompting on some datasets, with performance decreasing on the ChemProt dataset. Given ChemProt's more complex relation schema [23], these findings suggest that the effectiveness of few-shot prompting may depend on task complexity and how well the selected examples represent the target annotations **[NEED REFERENCE]**. This prompt-dependency is important for this project specifically as it provides a basis for developing and iterating on multiple Extraction Pipeline Iterations (EPI) during BioRED-based development, rather than treating prompt design as a single, fixed choice that remains unexamined.

This has a direct implication for how prompts are constructed in this project. Recent work testing LLM annotation directly against BioRED and other biomedical benchmark datasets found that providing the benchmark's own entity schema and annotation guidelines within the prompt substantially improved alignment with the benchmark's annotations compared to a prompt including only entity schema, excluding specific guidelines, across every model and dataset tested [24]. The study's authors discuss how this is not because of a failure of the underlying model's biomedical knowledge, but misalignment between the model's interpretation of the extraction task and the conventions that the benchmark's annotations follow [24]. This distinction is important for how benchmark-oriented prompting should be understood: including a benchmark's schema into a prompt is a matter of methodology, defining what the extraction system is instructed to extract, rather than overfitting to the benchmark. This motivates tuning each EPI developed in this project towards BioRED's specific entity and relation schema and guidelines, rather than relying on a generic extraction prompt as it aligns the pipeline's output with the specific ground truth it is developed against.

However, this choice also defined the scope of what Research Question 2 (RQ2) is able to evaluate. A pipeline based on a prompt that is tailored to a benchmark's schema it optimised to perform the specific extraction task the benchmark defines, rather than a domain-level general extraction task. Evaluating this project's final frozen pipeline against contemporary literature therefore investigates whether its fixed, BioRED-based task specification transfers across differing text distributions, not whether the extraction system generalises to other unseen annotation schemas. This project is therefore scoped as an investigation of this specific research question, rather than the development of a general-purpose, production-oriented extraction system. BioRED-geared prompting is therefore treated as the fixed task configuration that is used to answer RQ2.

### 2.2.3 Performance and Limitations

Despite this flexibility, LLM-based extraction introduces failures that are usually absent from, or less significant in fine-tuned supervised models **[NEED REFERENCE]**. The most widely discussed is drawing from domain knowledge: LLMs may produce extracted entities or relations that are plausible but not present in the supplied text [22]. This is particularly important for biomedical IE, where an extracted relation is only useful if it is grounded in the given literature, rather than drawn from existing background knowledge.

Problems also exist with LLM consistency and control: LLM outputs can vary across repeated runs on the same input [25], and extraction quality has been shown to be sensitive to changes in prompt wording or the specific examples chosen for few-shot prompting [16, 21]. Models may also fail to strictly follow a predefined relation schema, extracting relation types outside the intended ontology or omitting required output fields, which can complicate downstream pipelines that assume a fixed schema [25]. Together, this behaviour may create a precision-recall trade-off where more exhaustive extraction increases recall at the cost of precision, while more conservative prompting tends to do the opposite.

### 2.2.4 Structured Output

An important design consideration is the format in which extracted information is returned. Limiting LLM output to a structured, machine-readable format, such as JSON conforming to a predefined schema, rather than unstructured text generation is intended to make extraction results easily usable in downstream processing without requiring separate parsing steps [22]. For a pipeline that extracts entities and relations that are subsequently assembled into a KG (Section 2.3), this is an important methodological choice: KG construction requires each extracted entity and relation to be reliably identifiable as a discrete unit whereas unstructured free-text output requires an additional extraction from the output extractions. Structured output generation is therefore treated in this project as a component of the extraction pipeline itself, rather than a post-processing step.

### 2.2.5 Implications for Biomedical Information Extraction

In summation, the current literature provides a more nuanced view than the conclusion that LLMs perform interchangeably with specialised biomedical extraction models. LLMs reduce the dependence on task- and subdomain-specific fine-tuning and offer flexible, adaptable extraction across entity and relation types, which is an important practical advantage over supervised PLM pipelines discussed in Section 2.2.1. However, this flexibility comes with significant caveats: extraction performance that is sensitive to prompt construction and appears to vary across datasets and specific tasks while the generative aspect of these models comes with reliability concerns such generation of information not grounded in the source text, variable outputs, and schema non-compliance. For a pipeline intended to generalise from a benchmark corpus to unseen contemporary literature, this distinction is important: benchmark-evaluated precision and recall capture extraction accuracy on in-distribution text, but do not directly capture whether a frozen prompt-based pipeline remains robust when applied to literature it was not developed against. This motivates evaluating not only whether extraction quality transfers from BioRED to contemporary PubMed abstracts, but whether the specific points of failure discussed become more or less significant during generalisation.

## 2.3 Knowledge Graph Construction Using LLMs

### 2.3.1 Knowledge Graph Representation

A KG represents information as a set of entities, or nodes, connected by relations, or edges, with both nodes and edges, sometimes including additional information such as a confidence score, or source identifier **[NEED REFERENCE]**. The fundamental unit of a KG is the triple, which represents a single relationship between two entities. For example, the finding that warfarin is associated with intracranial haemorrhage can be represented as the triple:

$$(\text{Warfarin, Association, Intracranial Haemorrhage}).$$

A KG is fundamentally a collection of triples, with shared entities linking separate triples into a connected structure rather than a set of isolated relations.

This structure is well suited to biomedical information where individual findings are often reported in isolation within a single abstract but gain insight when connected across the literature. Representing extracted findings as a knowledge graph connects mentioned of the same entity across different papers into a single queryable structure. This can reveal clusters of relations and indirect connections that would not be apparent from a single abstract in isolation. For example, identifying an indirect association between two entities that are never mentioned together in the same source text, but are connected to each other through a mutual entity **[NEED REFERENCE]**.

This representation maps naturally onto the information extraction tasks introduced in Section 2.1. NER identifies the spans of text that become candidate nodes in the graph and NEN maps those candidate nodes to shared predefined identifiers, ensuring that semantically separate mentions of the same concept are consolidated into a single node rather than represented as duplicates. Finally, RE identifies the edges connecting these nodes. The insights KGs provide are precisely why NEN is a key component of the extraction pipeline: if duplicate mentions of a single concept were not resolved, biomedical concepts would become fragmented across several nodes, undermining the purpose that KGs are intended to provide.

### 2.3.2 LLM-Based Knowledge Graph Construction

Section 2.2.1 outlined how the recent LLM-driven work represents a broader shift in knowledge extraction from rule-based and modular pipelines towards adaptive generative frameworks [15]. In the context of KG construction, this shift changes the overall architecture of the entire extraction-to-graph system. A traditional, modular architecture treats NEN, NER, and RE as separate stages, each performed by a distinct model or tool, before the resulting entities and relations are assembled into a graph **[NEED REFERENCE]**. Although the exact ordering of these steps may differ, an example module workflow is:

$$\text{Text} \rightarrow \text{RE} \rightarrow \text{NER} \rightarrow \text{NEN} \rightarrow \text{KG}\ \textbf{[NEED REFERENCE]}.$$

An LLM-driven approach instead allows a single model to conduct RE and NER directly from text within one pipeline step, with a separate subsequent step conducting NEN before KG construction:

$$\text{Text} \rightarrow \boxed{\text{LLM joint NER + RE}} \rightarrow \text{NEN} \rightarrow \text{KG}.$$

This consolidation is possible as an LLM with a single adequate prompt can both recognise entity spans and the relations between them in text. Recent work evaluating end-to-end biomedical relation extraction using LLMs demonstrates the potential of joint-extraction [22] (?) and multi-extraction prompting has also been shown to support this within a single framework without requiring additional fine-tuning [25].

Furthermore, because the LLM's output is constrained to an easily-parsable schema, rather than free text, the extracted entities and relations are already in a form that does not require extra parsing or interpretation to be mapped onto graph triples. This is what allows the LLM-driven architecture to abstract several traditionally separate pipeline stages into fewer steps.

It is worth noting that LLMs can support KG construction beyond direct entity and relation extraction, including schema and ontology generation or refinement [15] **[NEED REFERENCE]**. This project primarily uses an LLM-based approach for structured information extraction, while also implementing LLM-assisted error analysis during iterative development of the extraction system. The graph schema and ontology itself, however, remain predefined, rather than being generated or refined by the LLM.

### 2.3.3 Ontology Integration and Entity Normalisation

Within KG construction, NEN provides the connection between entity mentions extracted from free text and established biomedical concepts. As discussed in 2.1.4, relying on separate nodes formed by unmapped string would fragment mentions of the same biomedical concepts across the graph. Mapping these mentions to ontology identifiers instead allows references to the same underlying concepts to be consolidated.

Biomedical ontologies and vocabularies address this by providing a fixed set of biomedical concepts, each associated with an established identifier, that extracted mentions can be mapped to **[NEED REFERENCE]**. NEN therefore acts as the step that allows a free-text span to be represented by an identifiable graph node.

Beyond identifier assignment, the ontology can also constrain the KG more broadly, restricting the extracted entities and relations to respective predefined, established sets **[NEED REFERENCE]**. This introduces a distinction between normalisation and ontology-based validation. While NEN maps an extracted span of text to a consolidated, established concept, validation can assess whether extracted entities and relations align with existing ontologies. Ontology integration can therefore support the consolidation and validation of the resulting KG **[NEED REFERENCE]**.

### 2.3.4 Challenges and Quality of LLM-Constructed Knowledge Graphs

Because a KG is assembled directly from the outputs of the extraction pipeline, errors made in the NER, RE, and NEN stages are not isolated from other stages, but instead pass into the resulting graph. For example, a false-positive entity or relation will become an incorrect node or edge, while false negatives would become missing edges. False-positive relations may be the more detrimental of the two errors, as the graph explicitly states them as knowledge rather than just excluding them **[NEED REFERENCE]**. Errors at the NEN stage, discussed in Section 2.3.3, have a similar effect: false-negative normalisation results in mentions of the same concept separated across duplicate nodes, while false-positive normalisation consolidates separate concepts into a single node **[NEED**

**REFERENCE]**.

Because of the connected nature of steps in an extraction pipeline system, KG quality is dependent on the quality of the original extraction. A KG can also be evaluated at the graph level for properties such as consistency, completeness, duplication, and compliance with its ontology or schema **[NEED REFERENCE]**.

## 2.4 Benchmark Datasets and Evaluation

### 2.4.1 Biomedical IE Benchmark Datasets

Progress in biomedical IE, has largely been measured against manually annotated benchmark corpora, which provide the ground truth against which NER and RE systems are trained and evaluated **[NEED REFERENCE]**. As annotation is costly and requires domain expertise, researchers typically constrain their annotations to particular biomedical entities, relations, or domains **[NEED REFERENCE]**, targeting a specific set of entity and relation types, rather than biomedical text in general **[NEED REFERENCE]**. Because of this, different benchmarks are not directly interchangeable as an extraction system's reported performance reflects the entities, relations, and schema of the specific benchmark it was evaluated on, rather than a measure of their general biomedical extraction ability. This benchmark heterogeneity was discussed in Section 2.2.1, where performance was shown to vary significantly between the GAD and ChemProt datasets.

This heterogeneity is stated briefly in table **[NEED REFERENCE]**, comparing the domain and entity/relation scopes of three benchmark datasets previously mentioned in this report.

Table 2.1: Comparison of biomedical relation extraction benchmark datasets.

| Dataset | Subdomain | Entity/Relation Type | Project Relevance |
|---|---|---|---|
| GAD [26] | Genetic association | Gene-disease association pairs: single binary relation type | Narrow, single-relation benchmark |
| ChemProt [23] | Chemical-protein interaction | Chemical and protein entities: multiple interaction subtypes | Benchmark with a more complex relation schema |
| BioRED [18] | General biomedical abstracts | Multiple entity types (e.g., gene, disease, chemical, species) and multiple relation types | Broader biomedical coverage |

This comparison demonstrates that the interpretation of a strongly-performing extraction system is dependent on the dataset it is built on: a model exhibiting strong results on the GAD dataset demonstrates a narrower scope of extraction strength than that of the BioRED dataset as the two differ substantially in terms of how many entity and relation types must be extracted and, therefore, understanding of context from biomedical literature. This difference motivates the choice of BioRED as the benchmark dataset used to develop the extraction pipeline in this project as a benchmark covering multiple entity and relation types is more representative of the broader task of general-purpose KG construction.

### 2.4.2 BioRED

BioRED, introduced in Section 2.2.1, is a manually annotated corpus of 600 PubMed abstracts, constructed to support document-level, multi-entry, multi-relation biomedical relation extraction **[NEED REFERENCE]**.

The corpus comprises six entity types: chemical entity (`ChemicalEntity`), disease or phenotypic feature (`DiseaseOrPhenotypicFeature`), gene or gene product (`GeneOrGeneProduct`), sequence variant (`SequenceVariant`), organism taxon (`OrganismTaxon`), and cell line `CellLine`. Eight relationship types connect these entities: association (`Association`), bind (`Bind`), comparison (`Comparison`), conversion (`Conversion`), cotreatment (`Cotreatment`), drug interaction (`Drug_`

`Interaction`), negative correlation (`Negative_Correlation`) and positive correlation (`Positive_Correlation`) [18].

In total, the corpus contains 20,419 entity annotations, mapped to 3,869 identifiers **[CHECK INFO]**, where, each entity mention is normalised to an established identifier in the annotation itself [18]. Relations are also labelled as describing either a new finding or existing background knowledge, allowing systems to distinguish novel findings from existing background information. As with the other benchmark corpora discussing in this review, BioRED is split into training, development, and test splits, enabling consistent comparison between systems evaluated on the dataset [18].

Because of these characteristics of the BioRED dataset, BioRED is well suited to the aims of this project. While other commonly-used biomedical RE benchmark datasets are constructed around a single entity or relation type, BioRED's wide scope of multiple entity types and relations and existing entity normalisation align closely with this project's general-purpose extraction task required for KG construction. This scope is the main motivator for its selection as the benchmark dataset used to iteratively develop an extraction pipeline in this project, ahead of evaluation against contemporary literature.

However, BioRED remains as fixed, manually annotated, benchmark-distributed corpus where annotations reflect the 600 abstracts sampled during construction and its entity and relation schema remains fixed rather than reflective of the full scope of biomedical concepts and relationships in contemporary literature. Because of this, while BioRED provides reliable ground truth for developing and evaluating an extraction pipeline, performance on the dataset does not necessarily reflect performance on literature published after the benchmark or on subdomains it was not designed to represent. BioRED is therefore treated in this project as a source of ground truth for extraction pipeline development, rather than as evidence of generalised extraction performance.

### 2.4.3 Extraction-Level Evaluation

Comparing biomedical IE systems, whether between the benchmark datasets in Section 2.4.1 or traditional and LLM-based approaches discussed in Sections 2.1.5 and 2.2, requires a consistent way of quantifying extraction quality. This is typically done by comparing a system's predicted entities or relations against a benchmark's manually annotated ground truth, classifying each prediction as a True Positive (TP), a correct prediction matching an annotation, a False Positive (FP), a predicted instance with no associated annotation, or a False Negative (FN), an annotation that the system failed to extract **[NEED REFERENCE]**.

Three derived metrics are then calculated from these counts: precision, the proportion of extractions that are correct, recall, the proportion of annotations that were correctly identified from the source text, and F1 score, the harmonic mean of precision and recall, commonly reported as a standalone summary figure, previously mentioned in Section 2.2.1 **[NEED REFERENCE]**.

Accuracy measures the proportion of predictions that are correct:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

However, biomedical extraction tasks usually involve many possible entity or relation candidates where true negatives can dominate. Accuracy can therefore be misleadingly high when a system extracts very little. In contrast, precision, recall, and F1 better represent the balance between over- and under-extraction discussed in Section 2.2.3 **[NEED REFERENCE]**. This framework applies both at the entity and relation levels. At the entity level, predictions are evaluated as span-type pairs, whereas relation-level predictions are made up of a pair of entities and the relation connecting them. Relation-level performance therefore also depends on the correctness of the underlying entity predictions **[NEED REFERENCE]**.

Additionally, the matching criteria used to determine whether an extracted entity or relation is correct should be noted. Under strict matching, a predicted entity span, type, or relation must exactly match an annotation to be counted as a true positive, with any boundary or string mismatch classifying the extraction as an FP and the ground truth annotation as an FN **[NEED REFERENCE]**. Using relaxed matching, some variation in the span, type, or relation is allowed, given the extraction still identifies the correct underlying concept **[NEED REFERENCE]**.

This distinction is imperative for LLM-based extraction specifically. As LLM outputs are generated rather than selected from a fixed set of labels, a predicted entity or relation may be semantically equivalent to an annotated instance while differing in exact expression – whether that is variation in exact string or its span boundary. Strict matching incorrectly classifies this variation as a genuine false negative, even when the underlying concept is correct. This motivates using a more relaxed or flexible matching approach when evaluating LLM-based extractions, illustrating the general limitations of strict matching for generative extractions. This project introduces supplementary evaluation approaches given the generative, variable nature of LLM-based extractions. The implementation of this approach is discussed in Section **[MORE DEPTH]**.

### 2.4.4 Limitations of Benchmark Evaluation

There are three distinct limitations that should be noted for benchmark evaluation, appearing at different parts of the evaluation pipeline.

The first is the dependence on a system's benchmark ground truth. Precision, recall, and F1 only reflect extraction performance relative the specific benchmark a system was built on as these metrics are only calculated directly against a fixed set of manual annotations. A predicted entity or relation that is correct in the broader scope of biomedicine can still be classified as an error if it does not meet the benchmark's specific annotation guidelines, entity boundaries, or relation definitions **[NEED REFERENCE]**. Performance figures therefore reflect a system's alignment with its benchmark's specific set of annotations, rather than its absolute extraction performance.

Secondly, building on the first limitation, as different benchmarks define different annotation schemas, F1 scores from systems build on different benchmark datasets are not necessarily directly comparable. A high F1 score on one benchmark and a lower F1 on another may not reflect a difference in extraction capability, rather a difference in schema complexity or annotation strictness between them. This has been observed previously in this literature review in Section 2.2.1 with the GAD/ChemProt datasets.

The third limitation, and most relevant to this project, is the external validity. Strong performance on a benchmark demonstrates only that a system performs well against its benchmark's specific collection of text and paper-level entities and relations. It does not inherently demonstrate that the same performance will hold against literature published after the benchmark was established or against subdomains and academic reporting practices that the benchmark did not sample.

In summation, these limitations do not invalidate the value of systems built upon benchmark datasets, rather their recognition helps to establish their scope. Benchmark evaluation remains essential as it provides controlled ground truth on which an extraction pipeline can be developed on. However, a system developed on a benchmark dataset cannot, by itself, demonstrate its generalisation to a separate distribution of literature. Addressing that gap is the motivation for this project's second research question.

## 2.5 Generalisation to Contemporary Biomedical Literature

### 2.5.1 Distribution Shift in Biomedical Literature

The limitations of benchmark evaluation discussed in Section 2.4.4 rely on the underlying assumption that a benchmark's fixed sample of text is representative of the literature a system will later be applied to. This assumption is important enough to warrant exploration as it provides the premise that the generalisation question this project investigates depends upon.

Biomedical literature is not static [2, 1]: new terminology and abbreviations become more commonly used, newly discovered entities and concepts are discovered, research focuses may shift over time, and reporting conventions may evolve **[NEED REFERENCE]**. This difference in the data a model or system is trained or tuned on and the data it encounters in real-world practice is generally called a distribution shift [27] and has been briefly discussed previously in Sections 2.1.4 and 2.4.4 in the observation that benchmark datasets are fixed while the literature they are intended to represent continues to grow.

A benchmark such as BioRED captures only a historical sample of biomedical literature, comprising abstracts available at the time of construction. As biomedical literature continues to grow evolve, later publications may shift away from the distribution that the benchmark represents, even where the general extraction task of identifying entities and relations from the biomedical abstracts remains the same. It should be noted that a distribution shift does not imply that a benchmark becomes invalid as a ground truth, but that it cannot be assumed that its representativeness of contemporary literature remains constant.

Literature exploring evidence for this caveat in the biomedical domain is relatively limited, as most existing work on distribution shifts over time has focussed more on general literature, rather than biomedical corpora specifically **[NEED REFERENCE]**. However, a recent study by Liu et al. (2025) measuring temporal effects across biomedical language models found that performance fell when models trained on data from one time period were evaluated on data from a later period, with the extent to which performance degraded depending on the specific task and the magnitude of the temporal shift [28]. Furthermore, NER exhibited greater sensitivity to shifts over other extraction tasks. This provides direct yet still scarce evidence that the distribution shift assumption associated with benchmark-based development is an important consideration within biomedical IE.

Together, this raises the uncertainty about whether performance established on BioRED will be maintained when the same extraction pipeline is applied to newer biomedical literature drawn from a newer distribution.

### 2.5.2 Generalisation of Biomedical Extraction Systems

The discussion of generalisation ability, or lack thereof, in Section 2.5.1 has established why contemporary biomedical literature may differ from a fixed benchmark's distribution. This section (Section 2.5.2) builds on the previous discussion to provide deeper review of how extraction performance differs when a system is evaluated outside the distribution it was developed on.

As discussed previously, BioREx, introduced in Section 2.1.5, was proposed specifically to improve generalisation through heterogeneous training across multiple biomedical RE datasets, rather than relying on a single benchmark's training data. Similarly, Sanger et al.'s 2025 evaluation of a single best-performing model across five different datasets was motivated by the same general problem: performance obtained on one biomedical dataset does not directly transfer to another [13]. Both examples indicate that cross-dataset generalisation in biomedical RE is a significant problem requiring explicit attention.

Further evidence comes from literature evaluating the same models across multiple biomedical relation classification datasets side by side. Purpura et al. (2024) compared fine-tuned Bidirectional Encoder Representations from Transformers (BERT)-based models (DistilBERT and PubMedBERT) against zero- and few-shot LLMs (Flan-UL2 and Llama 2) on three distinct biomedical relation classification datasets: GAD, the Compound-Protein Interaction (CPI) dataset, and a social-determinants-of-heath dataset [29]. Performance varied significantly between all three dataset for each model, with F1 scores on the GAD dataset substantially lower than those of the CPI dataset for the same models. This further supports the evidence from Section 2.2.1 that difference in extraction system performance exist between datasets, however, Purpura et al. (2024) demonstrates the same effect across all combinations of several different models and datasets.

Together, this evidence provides the basis of a more cautious view than the claim that LLMs generalise better. LLMs provide a suitable extraction system architecture for transferable tasks, given their ability to perform NER and RE without task-specific fine-tuning as discussed in Section 2.2.1. However, the existing literature does not conclude whether this flexibility translates into stable performance when a pipeline developed on one benchmark is applied to an unseen distribution. Specifically, whether an LLM-based extraction pipeline developed on BioRED maintains its performance when applied to newly assembled contemporary biomedical literature remains relatively unclear.

### 2.5.3 Research Gap and Implications

All previous sections of this literature review have established several main points of discussion: that automated biomedical IE is necessary given the scale of biomedical literature, that specialised PLMs achieve strong performance when fine-tuned on benchmark data, that LLMs offer easily adaptable extraction without task-specific fine-tuning while being sensitive to prompt, task, and dataset choice, that extraction errors can spread forward in into any KGs built subsequently, and that benchmark evaluation provides controlled, reliable ground truth but cannot by itself validate its performance on out-of-distribution data.

Together, these findings provide the basis for RQ2. Existing literature provides evidence of both distribution shift and variation in extraction performance across datasets, but does not, however, directly test the validity of LLM-based-extraction benchmark-to-contemporary transfer examined in this project. Specifically, the difference in performance observed by Liu et al. (2025) was measured across fine-tuned biomedical PLMs, rather than prompt-based LLMs operating without task-specific tuning, and Purpura et al.'s cross-dataset comparison, evaluates transfer between existing, already-annotated benchmarks rather than to independently sampled, contemporary literature. Whether an LLM-based extraction pipeline, developed and iterated against a fixed benchmark dataset, retains its extraction quality when applied unchanged to newly published biomedical literature therefore remains unclear.

This project addresses that gap directly. An LLM-based extraction pipeline is developed and iterated against BioRED, before being frozen and applied without further refinement to an independently assembled contemporary corpus, allowing its benchmark-level performance to be evaluated against a distribution it was not developed on.

# Chapter 3

# Methodology

## 3.1 Research Design

This study uses an experimental pipeline development design in which an LLM-based biomedical information extraction system is developed iteratively against a fixed benchmark corpus before being frozen and evaluated on independently sampled, contemporary biomedical literature. The study consists of four main stages: development of the extraction pipeline against BioRED, selection of a pipeline iteration, evaluation of the chosen iteration, and construction and evaluation of a knowledge graph from the resulting contemporary extractions.

The main methodological idea of this design was the separation between development and evaluation. BioRED served as a basis for training, development, and test splits where this project's iterative pipeline development, made up of a set of Extraction Pipeline Iterations (EPIs), was conducted solely against the training and development splits. Each EPI's prompt design and configuration was based on the previous EPI's performance and optional further error analysis – this approach began with an initial configuration (EPI 001), discussed further in Section 3.5, that established a baseline extraction pipeline using generic biomedical extraction instructions without BioRED-specific schema guidance. Pipeline development continued against the BioRED training split until performance plateaued, at which point the EPI configurations were frozen and no further prompt refinement or schema adjustment was performed. The frozen iterations were then evaluated against the BioRED development split, where their performance was compared with one another and the best performing EPI was selected. This distinction allows subsequent evaluation to be treated as a measure of the pipeline's performance, rather than further development.

The final EPI was then evaluated under two distinct stages. The first is benchmark evaluation, which measures the final EPI's extraction performance against BiORED's held-out ground truth, establishing its performance within the distribution it was developed on. The second stage was generalisation evaluation, which applies the same pipeline to an independently assembled corpus of contemporary PubMed literature that the pipeline was not iteratively developed on. As the pipeline was not adapted to perform specifically on the contemporary corpus, any significant degradation in performance between them can be attributed to a change in the dataset's literature distribution, rather than to any changes in the pipeline's configuration, allowing benchmark and contemporary corpus performance to be compared. Following extraction and normalisation on the contemporary corpus, the extracted entities and relations were assembled into a KG.

## 3.2 Datasets

### 3.2.1 BioRED Benchmark Corpus

The original BioRED corpus, as described in Section 2.4.2, was used as the benchmark corpus for extraction pipeline development and evaluation in this project. The corpus consists of 600 manually annotated PubMed abstracts, assembled and released by the National Center for Biotechnology Information (NCBI) and was obtained directly from the dataset's public GitHub repository. The BioRED-BC8 extension was also considered as it provides a more recent expert-annotated evaluation set: however, the original BioRED corpus was retained as the controlled benchmark as the study separately evaluates generalisation using literature independently retrieved through the PubMed Application Programming Interface (API), better reflecting the intended deployment pathway. BioRED-BC8 therefore represents a valuable alternative external evaluation dataset for future work.

The original version of the BioRED dataset divides 600 abstracts into a training split of 400 abstracts, a development split of 100 abstracts, and a test split of 100 abstracts. Each split was used for a distinct role in this study. The training split was used throughout iterative EPI development, providing the abstracts observed by the LLM in which all EPI configurations were developed on, the development split was then used to evaluate all frozen EPI configurations against one another, and the test spit was used exclusively for the final benchmark evaluation of the best-performing pipeline configuration and played no role in either pipeline development or EPI selection.

For each abstract, this project used the associated PubMed Identifier (PMID), title, and abstract text as the input passed to the extraction pipeline, together with the corresponding gold-standard annotations used as ground truth. These annotations consisted of each entity's type and text span and each relation's type. The following table summarises how each component of the BioRED dataset was used in this study:

Table 3.1: BioRED Split Summary

| BioRED Component | No. of Abstracts | Purpose |
|---|---|---|
| Training spit | 400 | EPI development and performance estimate |
| Development split | 100 | EPI selection (held-out within BioRED) |
| Test split | 100 | Final benchmark evaluation |

It should be noted that EPI refinement was conducted using a fixed 35-abstract subsample of the BioRED training split, with each final EPI then evaluated across the full 400-abstract training split. Section **[NEED REFERENCE]** discusses this methodological decision further.

BioRED's train, development, and test splits were each retrieved from the official BioRED GitHub repository as separate BioC-XML files. This process is discussed in Section 3.2.3.

### 3.2.2 Contemporary Biomedical Corpus

An independently-assembled corpus of contemporary biomedical PubMed abstracts was constructed as the external evaluation set used to investigate generalisation to out-of-distribution data, as mentioned in Section 2.5. This Contemporary Biomedical Corpus (CBC) included abstracts from papers published across 2024, 2025, and 2026 making up 282 total abstracts after duplication removal by PMID and removal of entries with missing abstract texts. It should be explicitly noted that this corpus was assembled entirely independently of BioRED and played no role in EPI development or selection.

"Contemporary biomedical literature" was defined through a fixed set of either PubMed research queries, designed to give broad coverage of the wider field of general biomedicine, rather than a single subdomain. Four of the eight related to individual biomedical entity categories, generally mirroring BioRED's own entity type (gene/protein, disease/disorder, drug/chemical, and variant/mutation), while the remaining four combined two entity categories within a single query (gene and disease, drug and disease, gene and drug, variant and disease). This query configuration was chosen to feasibly maximise the likelihood of retrieving abstracts that include entity relationships, rather than only single entity mentions. Each of the eight queries were combined with a publication year filter for all three publication years, giving 24 distinct query-year combinations with 15 articles requested per combination. The table below summarises each query category and associated term.

Table 3.2: Contemporary Biomedical Corpus Retrieval Query Summary

| Query category | Query terms |
|---|---|
| Gene/protein entity mention | `"gene" OR "protein"` |
| Disease/disorder entity mention | `"disease" OR "disorder"` |
| Drug/chemical entity mention | `"drug" OR "chemical"` |
| Variant/mutation entity mention | `"variant" OR "mutation"` |
| Gene-disease relation | `"gene" AND "disease"` |
| Drug-disease relation | `"drug" AND "disease"` |
| Gene-drug relation | `"gene" AND "drug"` |
| Variant-disease relation | `"variant" AND "disease"` |

Abstracts were retrieved from PubMed using the NCBI Entrez E-utilities, accessed through Biopython's `Bio.Entrez` module under a registered email address, as required by NCBI's usage policy. For each of the 24 query-year combinations an `esearch` request was used to return 15 matching PMIDs, article title, abstract text, journal title, and publication year were extracted. Author names, Digital Object Identifiers (DOIs), and keywords were not stored.

Across all query-year combinations, up to 360 records were requested in total: 15 results per query-year combination. After consolidating all returned results, duplicates were removed where PMIDs matched, along with any records with absent abstract text, leaving a final corpus of 282 abstracts. Beyond publication year, no sampling constraints were applied at this stage. A subset of the CBC was selected for manual annotation, in order to provide ground-truth entity and relation labels which the frozen extraction pipeline could be evaluated against. This subset comprised 50 randomly sampled abstracts from the 282 total entries. The annotation procedure and resulting metrics are discussed in Section 3.7.3.

### 3.2.3 Data Preparation

Although BioRED and the CBC were retrieved from different sources and through different processes, both required conversion into a structured, tabular representation before use by the rest of the pipeline. `pandas` DataFrames were used for both datasets, allowing later extraction and evaluation to be conducted easily over a tabular data, regardless of which dataset was used. BioRED was retrieved as raw BioC-XML and was parsed directly using `xmltodict`, whereas the CBC was retrieved via Biopython's `Bio.Entrez` module, which returns PubMed records already parsed into Python dictionaries, requiring no further XML-parsing step.

For BioRED, each of the three split's files was parsed with `xmltodict` into a Python dictionary before being converted into three separate `pandas` DataFrames per split: a DataFrame containing each abstract's PMID, title, and abstract text, an entity DataFrame containing one row per annotated entity mention, including its text span, entity type, and normalised identifier, and finally an entity relation DataFrame containing one row per annotated entity relation, including normalised identifiers of the two entities and their relation type.

For the CBC, the entire returned dictionary was iterated over to extract each paper's PMID, title, abstract text, and publication year. Where a paper had its abstract split into multiple segments, all segments were concatenated into a single string and duplicate records.

Relatively little additional preprocessing was conducted beyond this stage. For BioRED, no records were excluded during parsing and for the CBC, no records were excluded beyond removal of duplicates and papers that had missing abstracts. For both corpora, all abstracts and corresponding title text string remained untouched as retrieved as the extraction pipeline provides abstract and title text directly to the LLM as natural language, rather than preprocessed text. The output of this stage was therefore a set of standardised, pipeline-ready CSV files.

## 3.3 System Architecture

The processed datasets discussed in Section 3.2.3 formed the inputs to the extraction stage of the system. For BioRED, the resulting extractions were used for benchmark evaluation, whereas CBC predictions were subsequently passed through normalisation, validation, and KG construction. For each abstract, its title and text were passed to an LLM-based extraction step, which performed NER and RE jointly rather than as separate LLM API calls, returning structured entity and relation predictions. These predictions were parsed into the same tabular format used for BioRED's own annotations, allowing predicted and ground-truth extractions to be compared directly for subsequent evaluation. Figure **[NEED REFERENCE]** below summarises this data flow across the implemented pipeline architecture.

Extracted entity mentions then underwent entity normalisation, mapping each mention to its corresponding Concept Unique Identifier (CUI) in the Unified Medical Language System (UMLS), using SapBERT [30] to encode each mention and retrieve its nearest neighbour among UMLS concept vector embeddings by cosine similarity, before undergoing ontology-based validation, described in Section 3.6.3. The resulting normalised, validated entities and relations were subsequently converted into nodes and edges and assembled into the final KG.

The same extraction architecture was used across both BioRED and the CBC, differing only in the input data supplied and presence of gold-standard annotations, with BioRED additionally providing ground truth against which extraction-level performance could be measured, and the CBC providing only a manually annotated subset for this purpose, as described in Section 3.2.2. Although the pipeline's specific configuration changed across the EPI development process, the overall architecture described above remained consistent throughout.

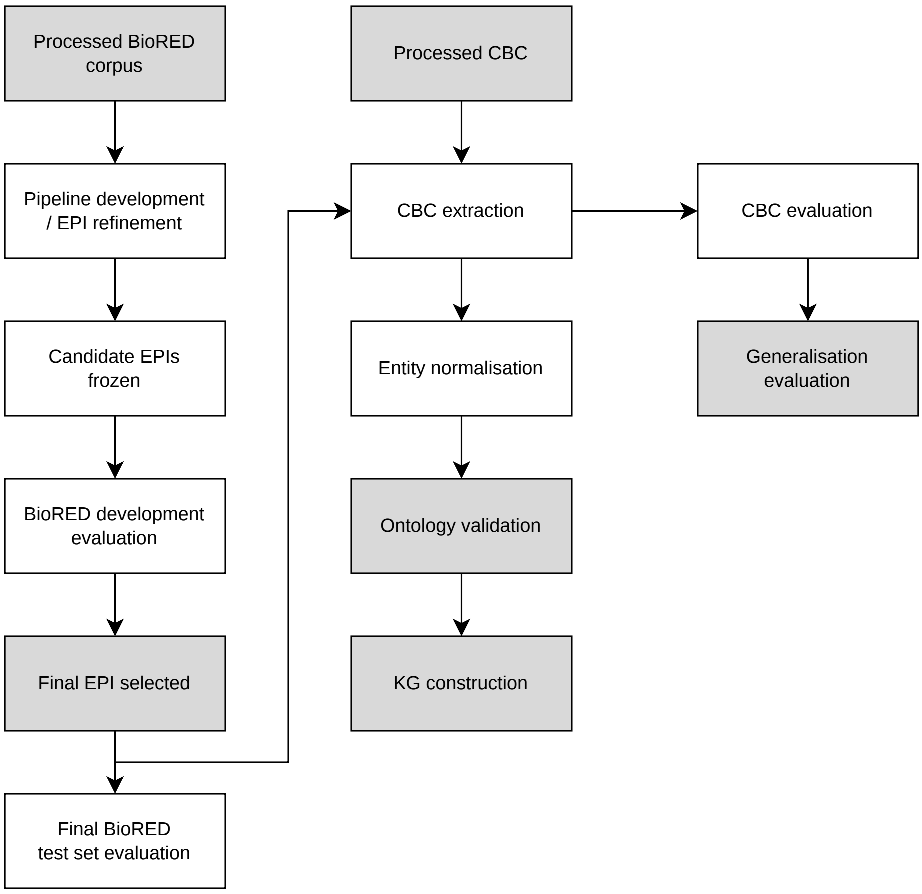


Figure 3.1: System architecture and data flow, from prepared corpus input through extraction, normalisation, and KG construction, to the three evaluation branches described in Section 3.7. Grey shading highlights key inputs, final stages, and outputs.

## 3.4 Information Extraction Pipeline

### 3.4.1 LLM Configuration

### 3.4.2 Prompt and Extraction Schema

### 3.4.3 Structured Output and Parsing

### 3.4.4 Entity Normalisation

## 3.5 Pipeline Development

### 3.5.1 Extraction Pipeline Iterations

### 3.5.2 Development and Model Selection

### 3.5.3 Final Pipeline

## 3.6 Contemporary Knowledge Graph Construction

### 3.6.1 Contemporary Corpus Extraction

### 3.6.2 Graph Construction

### 3.6.3 Ontology-Based Validation

## 3.7 Evaluation

### 3.7.1 Entity Evaluation

### 3.7.2 Relation Evaluation

### 3.7.3 Generalisation Evaluation

### 3.7.4 Knowledge Graph Evaluation

## 3.8 Cost Analysis

# Chapter 4

# Results

# Chapter 5

# Discussion

# Chapter 6

# Conclusion

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